

ADAM ROBERT PINES

Postdoctoral Scholar | Mitra and Brain Stimulation Laboratories | Stanford University

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EDUCATION

Ph.D., Neuroscience

August 2017 - August 2022

Dissertation: Layers of Maturation in Cortical Hierarchies

The University of Pennsylvania, Philadelphia, PA

Advisor: Theodore Satterthwaite, M.D., M.A.

Bachelor of Arts, *magna cum laude*, Psychology (Major), Biology (Minor)

August 2011 - May 2015

Loyola Marymount University, Los Angeles, CA

RESEARCH

Postdoctoral Scholar

Stanford University, Stanford, CA. PIs: Anish Mitra and Nolan Williams

May 2025 - Present

Stanford University, Stanford, CA. PI: Leanne Williams

August 2022 - April 2025

Graduate Student

August 2017 - August 2022

The University of Pennsylvania, Philadelphia, PA. PI: Theodore Satterthwaite

Clinical Research Coordinator

October 2015 - May 2017

Stanford University, Stanford, CA. PI: Leanne Williams

Research Volunteer

June 2015 - October 2015

Stanford University, Palo Alto, CA. PI: Amit Etkin

Research Assistant

September 2013 - May 2015

Loyola Marymount University, Los Angeles, CA. PI: Cheryl Grills

AWARDED FUNDING

Walter V. and Idun Berry Postdoctoral Fellowship (Stanford)

September 2025-

High Impact Technology Fund (Stanford)

September 2025-

Organization for Human Brain Mapping Merit Award

May 2025

Society of Biological Psychiatry Travel Award

April 2025

Psychiatry Trailblazing Trainee Award (Stanford)

February 2024 -

October 2025

School of Medicine Dean's Fellowship (Stanford)

July 2023 - July 2024

Ruth L. Kirschstein National Research Service Award (F31, NIMH)

February 2021 - August 2022

Jameson-Hurvich Travel Award for Behavioral Neuroscience (Penn)

June 2021

Achievement Award, Loyola Marymount University (LMU)

August 2011 - May 2015

PUBLICATIONS

Pines, A., Xhang, Z., Kolchalka, J., Akiki, T., Rajasekharan, D.*, Vesuna, S., Kauvar, I., Hack, L., Reneau, R., Siegel, J., & Williams, L. Psychedelics disrupt hierarchical propagations in the default mode network of humans and mice. *Proceedings of the National Academy of Sciences*. *Mentored Graduate student

Pines, A., Tozzi, L., Bertrand, C.*, Keller, A., Zhang, X., Whitfield-Gabrieli, S., Hastie, T., Larsen, B., Leikauf, J., & Williams, L. (2024). Psychiatric Symptoms, Cognition, and Symptom Severity in Children. *JAMA Psychiatry*. *Mentored Research Coordinator, now clinical psychology graduate student

Pines, A., Keller, A., Larsen, B., Bertolero, M., Ashourvan, A., Bassett, D., Cieslak, M., Covitz, S., Fan, Y., Feczko, E., Houghton, A., Rueter, A., Tapera, T., Vogel, J., Weinstein, S., Shinohara, R., Williams, L., Fair, D., & Satterthwaite, T. (2023). Development of Top-Down Cortical Propagations in Youth. *Neuron*.

Pines, A., Larsen, B., Cui, Z., Sydnor, V., Bertolero, M., Adebimpe, A., Alexander-Bloch, A., Davatzikos, C., Fair, D., Gur, R.C., Gur R.E., Li, H., Milham, M., Moore, T., Murtha, K., Parkes, L., Thompson-Schill, S., Shanmugan, S., Shinohara, T., Weinstein, S., Bassett, D., Fan, Y., & Satterthwaite T. (2022) Dissociable Multi-scale Patterns of Development in Personalized Brain Networks. *Nature Communications*.

Pines, A., Cieslak M., Larsen, B., Baum, G., Cook, P., Adebimpe, A., Dávila, D., Elliott, M., Jirsaraie, R., Murtha, K., Oathes, D., Piiwaa, K., Rosen, A., Rush, S., Shinohara, R., Bassett, D., & Satterthwaite, T. (2020) Leveraging multi-shell diffusion for studies of brain development in youth and young adulthood. *Developmental Cognitive Neuroscience*.

Pines, A., Sacchet, M., Kullar, M., Ma, J., & Williams, L. (2018) Multi-unit relations among neural, self-report, and behavioral correlates of emotion regulation in comorbid depression and obesity. *Scientific Reports*.

Zhang, X., **Pines, A.**, Stetz, P., Goldstein-Piekarski, A., Xiao, L., Lv, N., Lavori, P., Snowden, M., Venditti, E., Smyth, J., Suppes, T., Ajilore, O., Ma, J., & Williams, L. (2024). Adaptive cognitive control circuit changes associated with problem-solving ability and depression symptom outcomes over 24 months. *Science Translational Medicine*.

Cui, Z., **Pines, A.**, Larsen, B., Sydnor, V. J., Li, H., Adebimpe, A., Alexander-Bloch, A. F., Bassett, D. S., Bertolero, M., Calkins, M. E., Davatzikos, C., Fair, D. A., Gur, R. C., Gur, R. E., Moore, T. M., Shanmugan, S., Shinohara, R. T., Vogel, J. W., Xia, C. H., Fan, Y., & Satterthwaite, T. D. (2022). Linking Individual Differences in Personalized Functional Network Topography to Psychopathology in Youth. *Biological Psychiatry*.

Keller, A. S., **Pines, A.**, Shanmugan, S., Sydnor, V. J., Cui, Z., Bertolero, M. A., Barzilay, R., Alexander-Bloch, A. F., Byington, N., Chen, A., Conan, G. M., Davatazikos, C., Feczko, E., Hendrickson, T. J., Houghton, A., Larsen, B., Li, H., Miranda-Dominguez, O., Roalf, D. R., Perrone, A., Shinohara, R., Fan, Y., Fair, D., & Satterthwaite, T. D. (2023). Personalized Functional Brain Network Topography Predicts Individual Differences in Youth Cognition. *Nature Communications*.

Mehta, K.*, **Pines, A.**, Adebimpe, A., Larsen, B., Bassett, D., Calkins, M., Baller, E., Gell, M., Patrick, L., Gur, R.E., Gur, R.C., Roalf, D., Romer, D., Wolf, D., Kable, J., & Satterthwaite, T. (2023). Individual Differences in Delay Discounting are Associated with Dorsal Prefrontal Cortex Connectivity in Youth. *Developmental Cognitive Neuroscience*.

**Mentored Data Analyst, now neuroscience graduate student*

Muñoz Rodríguez, P.*, **Pines, A.**, Zhang, X., van Roessel, P., Mukunda, P., McCarthy, E., Williams L., & Rodriguez, C. (2025) Exploring the Effects of Cognitive Behavioral Therapy on Cognitive Control Circuit and Behavioral Task Performance in Hoarding Disorder. *Journal of Psychiatric Research*.

**Mentored Graduate student*

Williams, L., **Pines, A.**, Goldman Rosas, L., Goldstein-Piekarski, A., Lavori, P., Dagum, P., Wandell, B., Correa, C., Greenleaf, W., Suppes, T., Perry, L., Smyth, J., Lewis, M., Venditti, E., Snowden, M., Simmons J., & Ma, J. (2018). The ENGAGE study: Integrating neuroimaging, virtual reality and smartphone sensing to understand self-regulation for managing depression and obesity in a precision medicine model. *Behaviour Research and Therapy*.

Tozzi, L., Zhang, X., **Pines, A.**, Olmstead, A., Zhai, E., Anene, E., Chesnut, M., Holt-Gosselin, B., Chang, S., Stetz, P., Ramierz, C., Hack, L., Korgaonkar, M., Wintermark, M., Gotlib, I., Ma, J., & Williams L., (2024). Personalized brain circuit identify clinically distinct biotypes in depression and anxiety. *Nature Medicine*.

Keller, A. S., Sydnor, V., **Pines, A.**, Fair, D., Bassett, D., & Satterthwaite T., (2022). Hierarchical functional system development supports executive function. *Trends in Cognitive Sciences*.

Luo, A.*, Sydnor, V., **Pines A.**, [and 24 others] (2024). Functional Connectivity Development along the Sensorimotor–Association Axis Enhances the Cortical Hierarchy. *Nature Communications*.

**Mentored Graduate Student*

Keller, A. S., Mackey, A. P., **Pines. A.**, Fair, D., Hoffman, M.S., Salum, G., Barzilay, R., & Satterthwaite, T. (2022). Caregiver monitoring, but not caregiver warmth, is associated with general cognition in two large sub-samples of youth. *Developmental Science*.

Ashourvan, A., Shah, P., **Pines, A.**, Gu, S., Lynn, C., Bassett, D., Davis, K., & Litt, B. (2021). Pairwise maximum entropy model explains the role of white matter structure in shaping emergent co-activation states. *Nature Communications Biology*.

Murtha, K.*, Larsen, B., **Pines, A.**, Parkes, L., Moore, T. M., Adebimpe, A., Bertolero, M., Alexander-Bloch, A., Calkins, M. E., Davila, D. G., Lindquist, M. A., Mackey, A. P., Roalf, D. R., Scott, J. C., Wolf, D. H., Gur, R. C., Gur, R. E., Barzilay, R., & Satterthwaite, T. D. (2022). Associations between neighborhood socioeconomic status, parental education, and executive system activation in youth. *Cerebral Cortex*.

**Mentored Research Coordinator, now clinical psychology graduate student*

Li, B., Bailenson, J., **Pines, A.** Greenleaf, W., & Williams, L. (2017) A public database of immersive VR videos with corresponding ratings of arousal, valence, and correlations between head movements and self report measures. *Frontiers in Psychology*.

Jirsaraie, R., Gatavins, M., **Pines, A.**, Kandala, S., Bijsterbosch, J., Marek, S., Bogdan, R., Barch, D., & Sotiras, A. (2024) Mapping the Neurodevelopmental Predictors of Psychopathology. *Molecular Psychiatry*.

Hermosillo, R., Moore, L., Fezcko, E., Miranda-Domínguez, O., **Pines, A.**, Dworetzky, A., Conan, G., Mooney, M., Randolph, A., Graham, A., Adeyemo, B., Earl, E., Perrone, A., Carrasco, C., Uriarte-Lopez, J., Snider, K., Doyle, O., Cordova, M., Koiraal, S., Grimsrud, G., Byington, N., Nelson, S., Gratton, C., Peterson, S., Feldstein Ewin, S., Nagel, B., Dosenbach, N., Satterthwaite, T., & Fair, D. (2024). A Precision Functional Atlas of Personalized Network Topography and Probabilities, *Nature Neuroscience*.

Cieslak, M., Cook, P., He, X., [and 39 others, including **Pines, A.**] (2021). QSIPrep: An integrative platform for preprocessing and reconstructing diffusion MRI. *Nature Methods*.

Larsen, B., Cui, Z., Adebimpe, A., **Pines, A.**, Alexander-Bloch, A., Bertolero, M., Calkins, M. E., Gur, R. E., Gur, R. C., Mahadevan, A. S., Moore, T. M., Roalf, D. R., Seidlitz, J., Sydnor, V. J., Wolf, D. H., & Satterthwaite, T. D. (2021). A Developmental Reduction of the Excitation:Inhibition Ratio in Association Cortex during Adolescence. *Science Advances*.

Shah, P., Ashourvan, A., Mikhail, F., **Pines, A.**, Kini, L., Shinohara, R., Bassett, D., Litt, B., & Davis, K. (2019). Characterizing the role of the structural connectome in seizure dynamics. *Brain*.

Sydnor, V., Larsen, B., Bassett, D., Alexander-Bloch, A., Fair, D., Liston, C., Mackey, A., Milham, M., **Pines, A.**, Roalf, D., Seidlitz, J., Xu, T., Raznahan, A., & Satterthwaite, T. (2021) Neurodevelopment of the association cortices: patterns, mechanisms, and implications for psychopathology. *Neuron*.

Linguiti, S.*, Vogel, J., Sydnor V., **Pines, A.**, Wellman, N., Basbaum, A., Eickhoff, C., Eickhoff, S., Edwards, R., Larsen, B., McKinstry-Wu, A., Cobb Scott, K., Roalf, D., Sharma, V., Strain, E., Corder, G., Dworkin, R., & Satterthwaite T. (2023). Functional imaging studies of acute administration of classic psychedelics, ketamine, and MDMA: Methodological limitations and convergent results. *Neuroscience and Biobehavioral Reviews*.

**Mentored Research Coordinator, now medical student*

Vogel, J. W., Alexander-Bloch, A., Wagstyl, K., Bertolero, M., Markello, R., **Pines, A.**, Sydnor, V. J., Diaz-Papkovich, A., Hansen, J., Evans, A. C., Bernhardt, B., Misic, B., Satterthwaite, T., & Seidlitz, J. (2024). Deciphering the functional specialization of whole-brain spatiomolecular gradients in the adult brain. *Proceedings of the National Academy of Sciences*.

Shanmugan, S., Seidlitz, J., Cui, Z., Adebimpe, A., Bassett, D., Bertolero, M., Davatzikos, C., Fair, D., Gur, R. E., Gur, R. C., Larsen, B., Li, H., **Pines, A.**, Raznahan, A., Roalf, D., Shinohara, R., Vogel, J., Wolf, D., Fan, Y., Alexander-Bloch, A., & Satterthwaite, T. (2021). Sex differences in functional topography of association networks. *Proceedings of the National Academy of Sciences*.

Yang, H., Wu, G., Li, Y., Xu, X., Ma, Y., Chen, R., **Pines, A.**, Xu, T., Sydnor, V., Satterthwaite T., & Cui, Z. Connectional axis of individual functional variability: Patterns: structural correlates, and relevance for development and cognition (2025). *Proceedings of the National Academy of Sciences*.

Keller, A., Moore, T., Luo, A., Visoki, E., Gatavins, M., Shetty, A., Cui, Z., Fan, Y., Feczko, E., Houghton A., Li, H., Mackey, A., Miranda-Dominguez, O., **Pines, A.**, Shinohara, R., Sun, K., Fair, D., Satterthwaite, T., & Barzilay, R. (2024). A general exposome factor explains individual differences in functional brain network topography and cognition in youth. *Developmental Cognitive Neuroscience*.

Richie-Halford, A., Cieslak, M., Ai, L., Caffarra, S., Covitz, S., Franco, A., Karipidis, I., Kruper, J., Milham, M., Avelar-Pereira, B., Roy, E., Sydnor, V., Yeatman, J., **The Fibr Community Science Consortium**, Satterthwaite T., & Rokem, A. (2022). An analysis-ready and quality controlled resource for pediatric brain white-matter research. *Scientific Data*.

Baller, E. B., Valcarcel, A. M., Adebimpe, A., Alexander-Bloch, A., Cui, Z., Gur, R. C., Gur, R. E., Larsen, B. L., Linn, K. A., O'Donnell, C. M., **Pines, A.**, Raznahan, A., Roalf, D. R., Sydnor, V. J., Tapera, T. M., Tisdall, M. D., Vandekar, S., Xia, C. H., Detre, J. A., Shinohara, R. T., & Satterthwaite, T. D. (2022). Developmental coupling of cerebral blood flow and fMRI fluctuations in youth. *Cell Reports*.

Xia, C., Barnett, I., Tapera, T., Cui, Z., Moore, T., Adebimpe, A., Rush-Goebel, S., Piiwaa, K., Murtha, K., Linguiti, S., Leibenluft, E., Brotman, M., Martin, M., **Pines, A.**, Calkins, M., Roalf, D., Wolf, D., Bassett, D., Lydon-Staley, D., Baker, J., Ungar, L., & Satterthwaite T. (2022). Mobile Footprinting: Linking Individual Distinctiveness in Mobility Patterns to Mood, Sleep, and Brain Functional Connectivity. *Neuropsychopharmacology*.

Zhao, S., Su, H., Cong, J., Chen, P., Wu, G., Li, Y., Fan, Q., Ma, Y., Xu, X., Yang, H., Li, H., **Pines, A.**, Chen, R., & Cui, Z. (2024) Hierarchical individual variation and socioeconomic impact on personalized functional network topography in children. *BMC Medicine*.

Butler, E., Samia, N., Mejia, M., Pham, D., **Pines A.**, & Nusslock, R. Sex differences in response to violence: Role of salience network connectivity and expansion on depression.

Keller, A.S., Sun, K. Y., Francisco A., [and 24 others, including **Pines, A.**] Reproducible Sex Differences in Personalized Functional Network Topography in Youth.

SUBMITTED

Jarukasemkit, S., Lenzini, P., Chen, A., Glasser, M., Hamilton, K., Harms, M., Luo, L., Myers, M., **Pines, A.**, Reid, E., Li, L., Zavaliangos-Petropulu, A., Zhang, J., Whitfield-Gabrieli, S., Williams, L., Sheline Y., Bijsterbosch, J. HARMONY: A large-scale harmonized neuroimaging dataset for research on anxious misery.

Meisler, S., Cieslak, M., Bagautdinova, J., [and 48 others, including **Pines, A.**] Highly replicable multisite patterns of adolescent white matter maturation.

Arns, M., Kaloiani, I, Wada, M, Dannawi, N., Ehrie, J., Seilheimer, R., Scangos, K., Austelle, C., Coetzee, J., Reid, B., **Pines, A.**, Buch, V., Halpern, C., & Rolle, C. Intracranial Stimulation induced Heart-Brain Coupling Confirms Anterior Cingulate Involvement in Top-Down Heart Rate regulation.

Mikolas, P., **Pines A.**, Zavaliangos-Petropulu, A., Al-Sharif, N., Norris, V., Lenzini, P., Akiki, T., Morfini F., Makhoul, W., Verhees, F.G., Ritter, P., Whitfield-Gabrieli, S., Sheline, Y., Biljsterbosch, J., Williams, L., & Narr, K. Large language models assisted classification of symptom domains across anxious misery disorders.

Zhou, D., Kim, J. Z., **Pines, A.**, Sydnor, V. J., Roalf, D. R., Detre, J. A., Gur, R. C., Gur, R. E., Satterthwaite, T. D., & Bassett, D. S. Compression supports low-dimensional representations of behavior across neural circuits.

Li, Y., Tan, L., Duan, Y., Xu, X., Xu, H., Yu, M., Jia, L., Li, Z., Zhao, C., Chen, Q., Larsen, B., **Pines, A.**, Wang, T., Chen, R., & Cui, Z. Normative executive function development reveals age-varying mental health associations in youth.

TEACHING

<i>Precision functional mapping of human brain networks</i> , Stanford University	January 2026
<i>Adaptive flexibility in the pace of brain development</i> , San Quentin Rehabilitation Center	June 2025
<i>Music and the brain</i> , University of San Francisco	March 2025
<i>The organization of neurocognition</i> , University of Kansas	November 2024
<i>Precision Mental Health and Wellness trainee workshop series</i> , Stanford University	Spring-Summer 2024
<i>Hierarchical neurodevelopment</i> , University of Kansas	April 2024
<i>Hierarchical cognition</i> , University of Kansas	April 2024
<i>Contemporary MRI processing pipelines for neuroscience</i> , University of Kansas	March 2024
<i>Hierarchical neuroaesthetics</i> . University of San Francisco	October 2022
<i>Introduction to the brain and behavior</i> . Teaching Assistant, University of Pennsylvania	Fall 2019

TALKS

Development of top-down cortical propagations. Organization for Human Brain Mapping **June 2025**
Optical flow for capturing hierarchical cortical activity propagations. Gradients of Brain Organization **June 2025**
The relationship between psychiatric symptoms and cognition depends on symptom severity,
Chinese Institute for Brain Research **October 2024**
Hierarchy-traversing cortical activity in neurodevelopment and psychedelics,
Washington University. in St. Louis **June 2024**
Development of top-down cortical propagations, University of Maryland, College Park **March 2024**
Co-existence of negative and positive associations between cognition and intergenerational psychiatric symptoms reveal necessity of socioeconomic and clinical enrichment, Boehringer-Ingelheim Idea Club **March 2024**
Cortical functions and psychopathologies in adolescence. Northeastern University **June 2023**
Hierarchical neurodevelopment, The University of Sydney **June 2023**
Development of top-down cortical propagations, Stanford University **January 2023**
Multimodal approaches to delineating neurocognitive development, Stanford University **January 2022**
Dissociable multi-scale patterns of development in brain networks, University of Minnesota **February 2021**

SERVICE

California State Science and Engineering Fair, Panel Judge **April 2023 & 2024**
Mind Center for Outreach, Research, and Education Fellowship Mentor **Summer 2018**
Guadalupe Homeless Project Volunteer **Fall 2012-Spring 2013**

WORK EXPERIENCE

American Council on Exercise Certified Personal Trainer

Burns Recreation Center, Westchester, CA **August 2013 - May 2015**
24 Hour Fitness, Mountain View Sport, Mountain View, CA **June 2015 - October 2015**

REVIEWS

Biological Psychiatry, Biological Psychiatry: Cognitive Neuroscience and Neuroimaging, Biological Psychiatry: Global Open Science, BMJ Global Health, BMC Medicine, CNS Neuroscience and Therapeutics, Developmental Science, Human Brain mapping, Imaging Neuroscience, Molecular Psychiatry, Neuroimage, Nature Communications, Nature Medicine, Nature Mental Health, Journal of Affective Disorders, The Journal of Child Psychology and Psychiatry, npj Science of Learning, PLOS Biology, PLOS One, Proceedings of the National Academy of the Sciences, Science Advances, Science Bulletin